

Switch Adapted Zuru XShot Motorized Rage Fire

MAKER GUIDE

Required Components

1. 	2. 	BOM 1. Zuru XSHOT Motorized Rage Fire (Walmart, Amazon, Canadian Tire) 2. (2x) 3.5 mm Mono Jack and Ring 3. 22 AWG Wire 4. (6x) AA Batteries 5. Toggle switch Refer to the Tool and Component List for purchasing details.
3. 	4. 	5. 

Required Tools

- Small Phillips Screwdriver 3mm Diameter
- Ruler
- Wire strippers
- Soldering iron and solder
- Drill and 1/4" drill bit

Required Personal Protective Equipment (PPE)

- Safety glasses



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Files available at https://github.com/makersmakingchange/Switch_Adapted_Toys

Switch Adapted Zuru XShot Motorized Rage Fire

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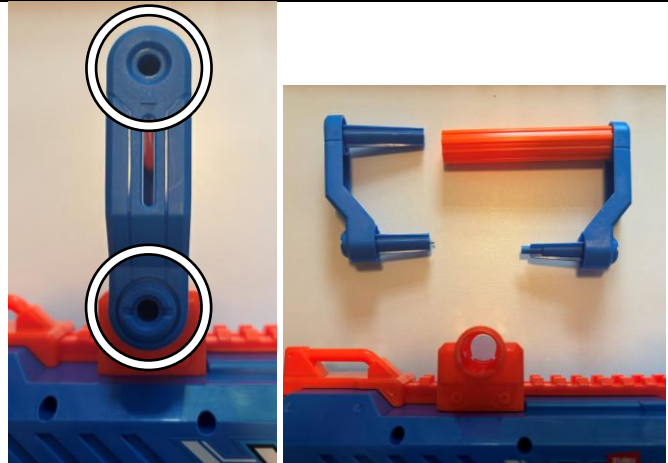
Assembly Instructions

1. Carefully remove the toy from its packaging without damaging it (we will be repackaging the toy after adapting). Carefully open up the cardboard, and then on the backside of the toy, twist off the plastic pieces holding the toy to the cardboard (shown circled in the photo).



2. Remove the two screws from inside the handle at the top of the toy, shown circled here, then remove the handle.

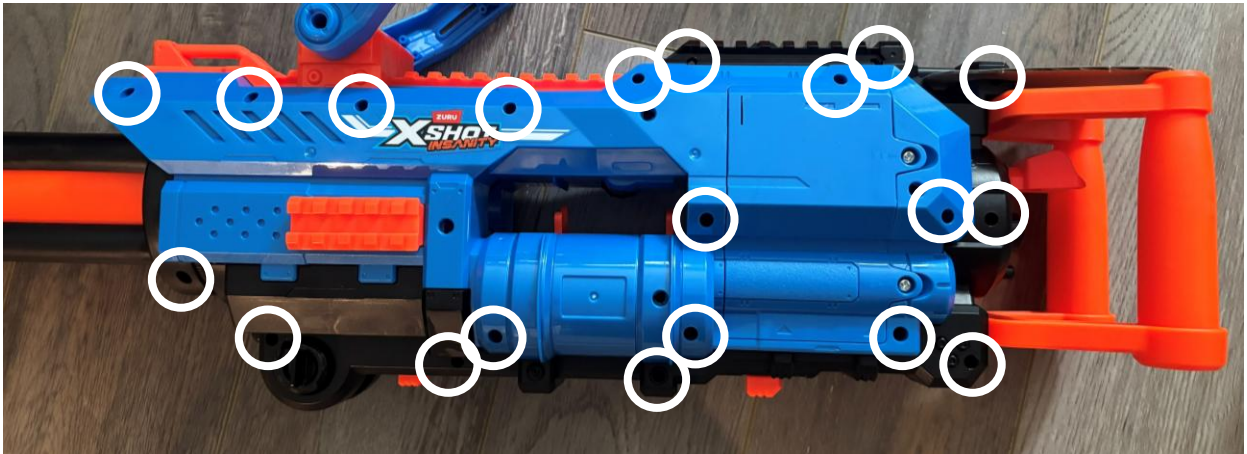
*Note: keep screws, and make sure not to lose them



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3. Locate and remove the 20 screws along the body of the toy.

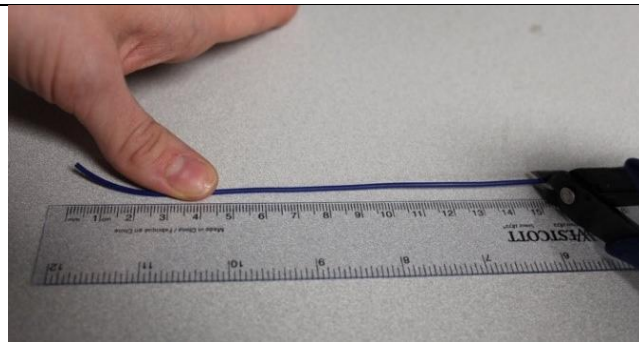
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4. Carefully separate the two halves of the toy, making note of any pieces that move or fall out.



5. Cut 6 pieces of wire, each 12 cm long.



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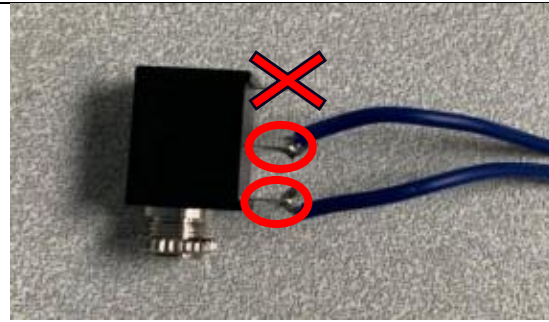
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6. Use wire strippers to strip approximately 0.5 cm off each end of the wires



7. Solder two of the wires onto the front two tabs of the first mono jack, closest to the jack input.

Repeat with the second mono jack.

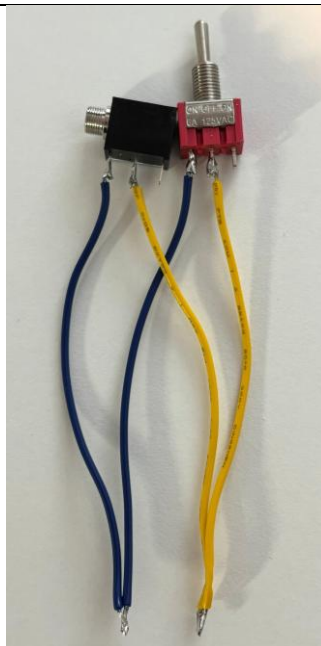


8. Solder two of the wires onto the left and middle tabs of the toggle switch.



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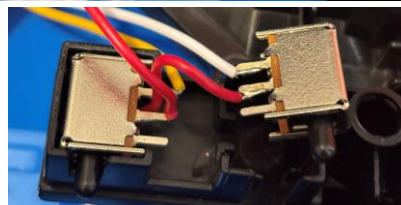
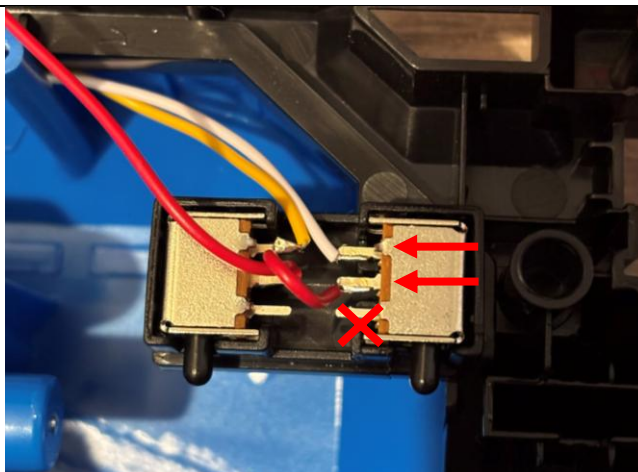
9. Take one wire from the toggle switch and one wire from one of the mono jacks and twist the ends of the wires together and solder them. Then repeat with the remaining two wires.



10. On the other mono jack (that is not soldered to the toggle switch), tin the other ends of the wires by adding a small amount of solder to each wire.



11. Take the mono jack and toggle switch that are soldered together and solder them in parallel with the existing wires connected to the switch on the right, in the positions shown (the two legs furthest from the switch). Do this by carefully soldering the new wires on top of the existing wires, being careful not to break the original connection. Each of the two solder joints will now have 3 wires: the original wire, the mono jack, and the toggle switch.

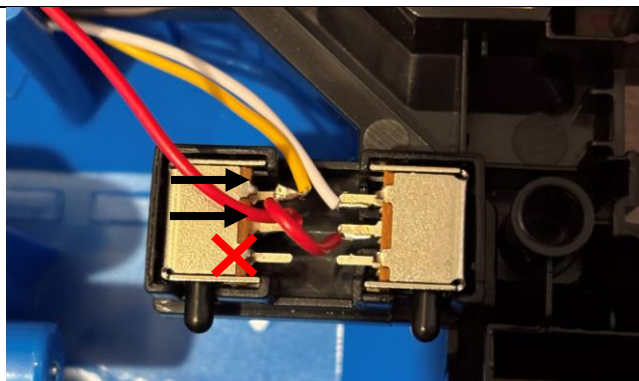


Note: if it is easier, you can carefully lift/pry the switch out of the plastic enclosure (as shown in the second photo) to give you more space to access the solder spots.

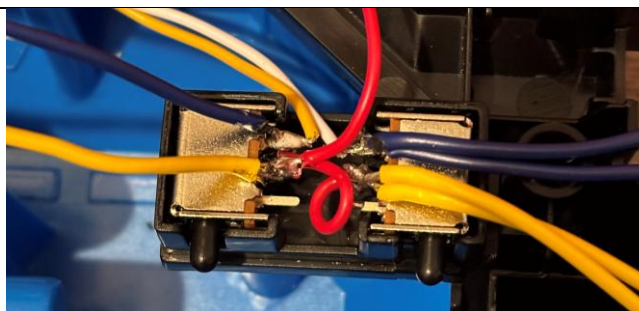
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12. Next, take the other mono jack and solder it in parallel with the existing wires connected to the switch on the left, in the positions shown (the two legs furthest from the switch). Do this by carefully soldering the new wires on top of the existing wires, being careful not to break the original connection. Each of the two solder joints will now have 2 wires: the original wire and the mono jack.



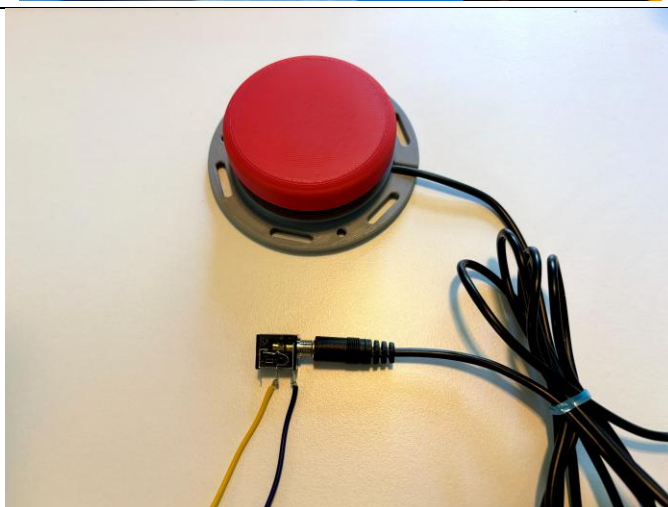
13. Once all connections have been soldered, it should look something like this photo, with the switch on the left soldered to a mono jack, and the jack on the right soldered to both a mono jack and the toggle switch.



14. It is now time to test the toy. Insert batteries into the battery compartment and assistive switches into the mono jacks. Test to make sure that the toy can be controlled using the original switch, the assistive switches, and the toggle switch. Do not use foam darts while testing, just listen for the motor.

Notes:

- The toggle switch and the mono jack connected to the switch on the right both control the “charge up” motor. You will hear the motor but will not see things move. Test these one at a time.
- The mono jack connected to the switch on the right should control the “shoot” function, the barrel at the front of the blaster will spin and the wheel that feeds the darts will also spin.

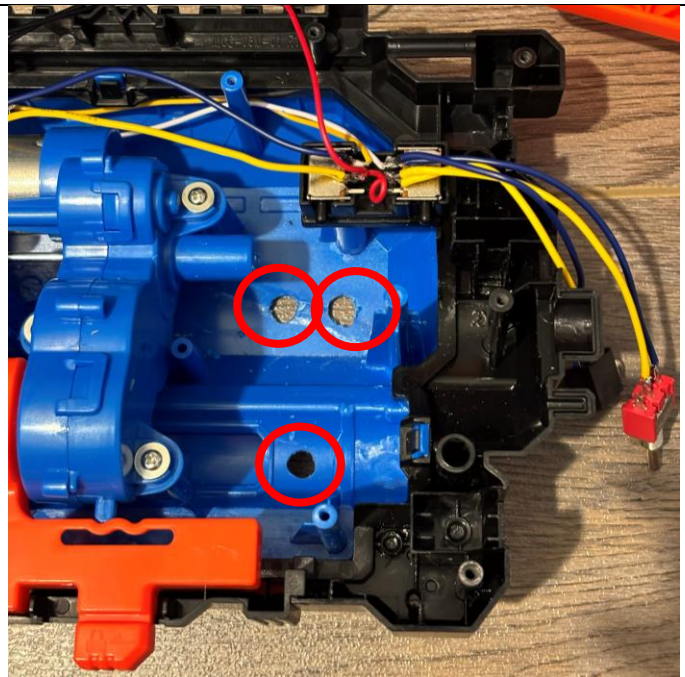


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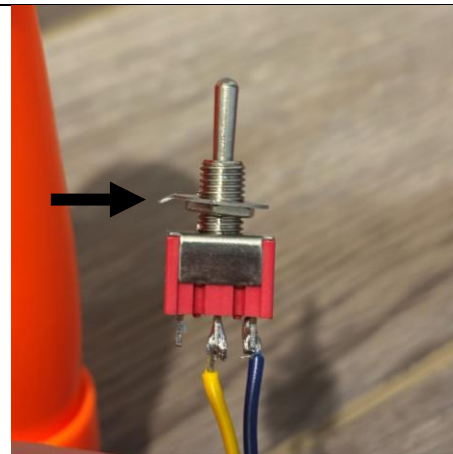
Troubleshooting:

- If the toy does not turn on, with any of the activation methods, please check the soldered connections.
- If the toy does not turn off when the assistive switch is released and the toggle switch is turned off, the connection is shorted. Check for areas of touching solder and separate using a soldering iron.

15. Using a $\frac{1}{4}$ " drill bit, drill 3 holes in the positions shown on the right. Remove any loose plastic.

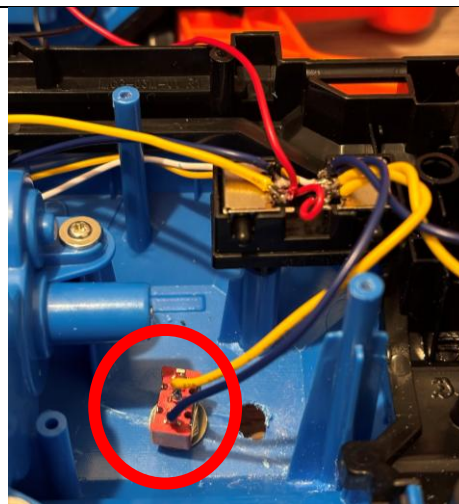


16. Take the toggle switch and put on one of the nuts and then the washer with a small tab on it (shown below), with the tab angled down or towards the wires, as shown on the right.

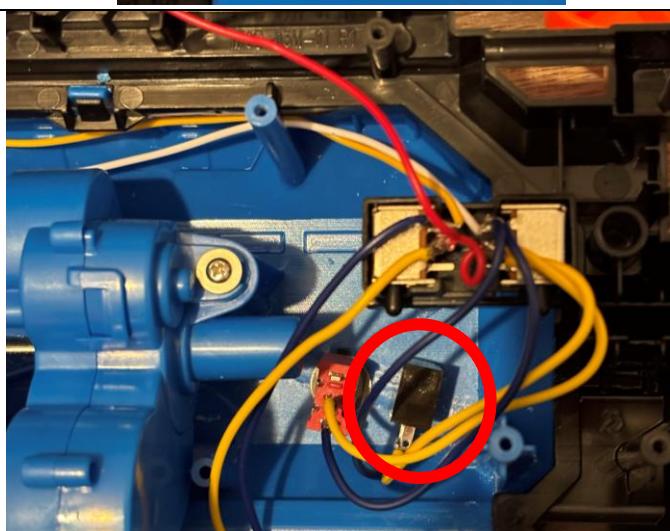


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17. Put the toggle switch through the upper left hole. Then from the outside of the toy, add the other washer with barbs and the second nut (shown below).



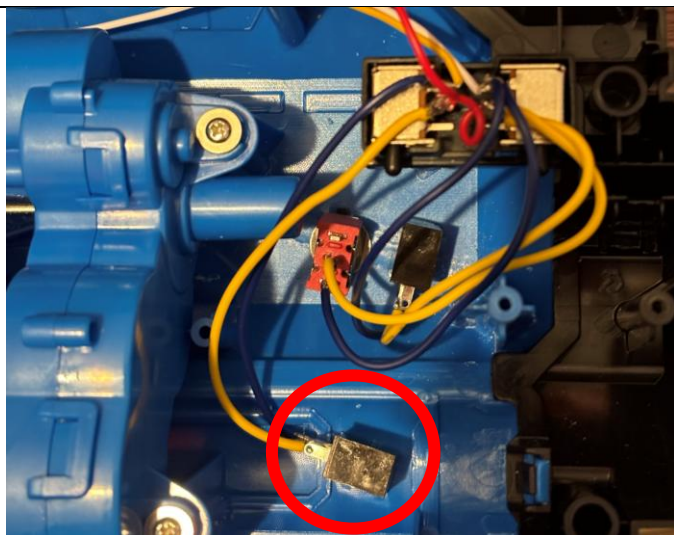
18. Now, take the mono jack that is soldered to the right switch, the same switch as the toggle switch. If the retaining ring is on the jack, remove it and push the mono jack through the hole beside the toggle switch from the inside. Reinstall the retaining ring on the outside of the toy and tighten to secure.



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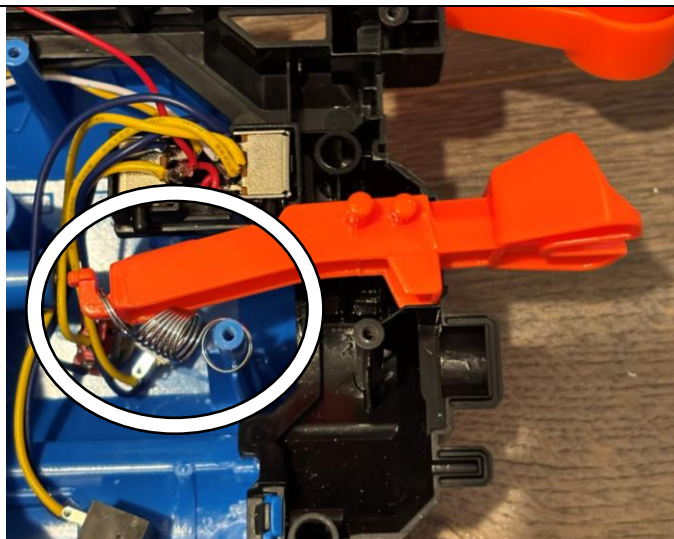
19. Take the remaining mono jack, if the retaining ring is on the jack, remove it and push the mono jack through the bottom hole from the inside. Reinstall the retaining ring on the outside of the toy and tighten to secure.



20. Now we are going to reassemble the toy, start by reinstalling any pieces that moved or fell out when taking the toy apart.

The trigger piece will need to be put back into place, with the spring on the end of the part and on the blue post nearby, as shown.

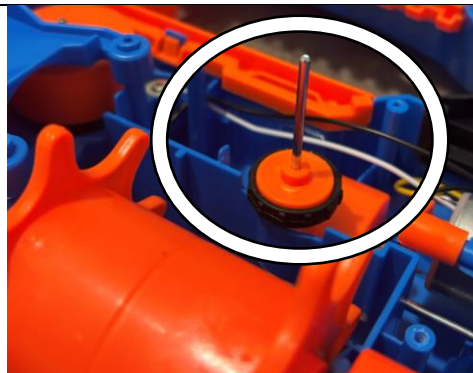
Check for any other pieces that may have fallen out or out of place, by referencing the second photo shown here.



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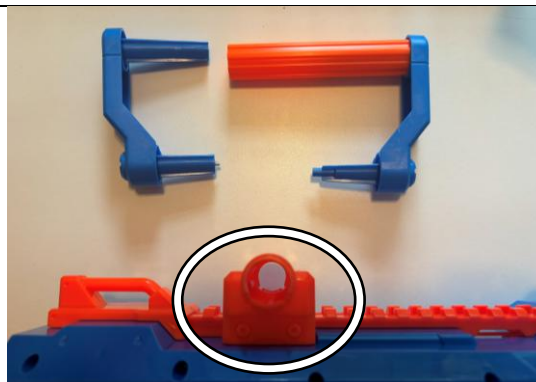
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21. Now, carefully put the two halves of the toy back together, making sure everything is lined up as you do so. In particular, make sure the part shown in this photo is lined up with the hole on the other side, as this can be tricky.
22. Once the two halves are together, reassemble the toy using the screws set aside earlier.



23. Now, reassemble and reinstall the handle to the top of the toy. Put the half of the handle that is still attached to the orange part through the orange hole on the top of the toy (circled in the photo), then take the other half of the handle and insert it into both the hole on the blaster and the other half of the handle.

Screw the two screws back in.



24. Test the toy again. If it is still working properly, test again with the darts this time. If it shoots the darts properly and works as expected, repackage the toy and the adapted toy is complete!

